

Heritage Treasures: An In-Depth Analysis of
UNESCO World Heritage Sites in Tableau

TEAM ID	SWTID-2026-4638
PROJECT TITLE	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
MAX MARKS	3 MARKS

6.2 – Data Connectivity

Data connectivity is the technical foundation of the "Heritage Treasures" project, dictating how raw international datasets are ingested, linked, and optimized within Tableau Desktop 2025.3.1. The connection workflow began by configuring a native file data connection to the raw UNESCO comma-separated value repositories. Instead of relying on rigid, traditional database joins that flatten data tables and duplicate rows, the project utilizes Tableau's advanced logical layer to establish flexible "Noodle" relationships using unique site identification keys as the common attribute. To maximize performance and ensure fast interface interactions, the data connection style was shifted from a live query mode to an optimized Data Extract configuration. Tableau's Hyper data engine converts the flat source files into an in-memory columnar database structure, explicitly indexing the spatial coordinates, historical years, and category strings. This extract configuration minimizes processing overhead by storing calculated values directly within the columns, allowing the system to run complex level-of-detail (LOD) calculations and multivariable filters without needing to re-scan the original text files. Additionally, geographic role attributes were explicitly assigned during the connectivity setup phase. The raw latitude and longitude decimal fields were mapped to native Tableau geographic fields, allowing the system to instantly recognize international country names and regional boundaries. This robust data model establishes a highly flexible analytical foundation. It enables the dashboards to handle deep spatial and chronological queries smoothly.